On Variance Reduction in Learning Mean Flows

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Abstract

Mean Flows enable distillation-free one-step generation by learning two-parameter average-velocity fields from noise to data distributions, but their training is notorious for non-decreasing loss and high gradient variance. We establish a theoretical framework that explains the phenomenon through the lens of *variance reduction with a control variate*. We identify two distinct statistical roles for the conditional velocity $\boldsymbol{v}_{\text{cond}}$ in training: as an unbiased regression target and as the Monte Carlo control variate for the tangent in the Jacobi-vector-product (JVP), but with a statistically suboptimal coefficient. The latter inflates per-step gradient variance and produces the non-decreasing loss. We derive the optimal tangent-mixing coefficient β^* in closed form and show that a family of concurrent works correspond to different practical realizations of the same optimum. To validate the framework, we instantiate the $\beta = 1$ corner via an EMA-derived proxy with a flow-matching anchor and run a controlled β-sweep ($\beta \in \{0, 0.25, 0.5, 1\}$) at both toy and DiT-B/4 / ImageNet-256 scale. The sweep recovers the predicted bias-variance ordering at every matched-step checkpoint, with $3\times \sim 5\times$ gradient-variance reduction and up to 54% SW_1 improvement on 2-D benchmarks, and $\approx 5\times$ reduction in gradient noise ratio at DiT scale. The same DiT-scale measurement also exposes a quantitative *FID-vs-MSE landscape mismatch*: the FID-axis bias amplification across $\beta = 0.5/\beta = 1$ (~ 1.4:1) is sublinear in the gradient-MSE bias coefficient (1:4), so the FID-optimal β shifts beyond the gradient-MSE interior minimizer to the unbiased corner $\beta = 0$.

1 Introduction

Deep generative models that leverage neural networks to parametrize and sample from unknown data distributions have achieved remarkable success [1, 2, 3, 4, 5]. Among these models, flow-based generative models [6, 7, 8, 9, 10, 11, 12] learn a continuous-time transformation between a simple prior and the data distribution. Conditional flow matching [10, 11] and stochastic interpolants [12, 13] further enable simulation-free training by regressing a velocity field, achieving sample quality competitive with diffusion models [1, 14, 2]. However, integration at inference time remains a challenge due to its computational costs and numerical instability, motivating works on *one-step generation*, including progressive distillation [15], distribution matching [16, 17], consistency models [18, 19], and flow map matching [20]. Nevertheless, most of them require a pre-trained distillation teacher.

Mean Flows [21] provide a *distillation-free* alternative by learning a two-parameter average velocity field $\boldsymbol{u}(\boldsymbol{x}, r, t)$ through regressing a total-derivative identity between the average and instantaneous velocity. In principle, this identity yields exact self-supervision. In practice, however, training is plagued by *non-decreasing loss* and prohibitively *high gradient variance* [22, 23]. A plethora of concurrent works have proposed remedies that target different mechanisms. AlphaFlow [22] attributes slow convergence to gradient conflict between flow-matching and total-derivative consistency components and proposes an α-curriculum. Improved MeanFlow [23] replaces the conditional-velocity tangent with a learned marginal-velocity head. Re-MeanFlow [24] preprocesses the data with rectified-flow straightening to reduce trajectory curvature. Terminal Velocity Matching [25] sidesteps the spatial Jacobian by differentiating with respect to the terminal time. Functional Mean Flows [26] extend MeanFlow to Hilbert spaces and study marginal-conditional consistency for functional data. Each fix is empirically effective in isolation, but there is no theory that explains *why* the original objective is unstable or what these fixes have in common.

In this paper, we establish our theory around a single research question:

Why do stochastic tangents destabilize learning Mean Flows?

We answer this question through the lens of *variance reduction*. Specifically, we identify that the conditional velocity $\boldsymbol{v}_{\text{cond}} = \boldsymbol{x}_1 - \boldsymbol{x}_0$ plays two statistically distinct roles in the MeanFlow loss. As the regression target, it is an *unbiased* estimator of the marginal velocity $\boldsymbol{v}$. As the tangent inside the Jacobian-vector product (JVP) of the total-derivative identity, however, the same $\boldsymbol{v}_{\text{cond}}$ acts as a Monte Carlo *control variate* for the inaccessible marginal velocity, with a statistically suboptimal coefficient. The bias-variance trade-off generated by this substitution governs gradient variance through the quadratic form $\boldsymbol{g}^\top \boldsymbol{J} \Sigma_{\boldsymbol{v}'} \boldsymbol{J}^\top \boldsymbol{g}$, where $\boldsymbol{g} = \nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}}$ and $\boldsymbol{J}$ is a spatial-Jacobi factor. The conditional fluctuation $\boldsymbol{v}_{\text{cond}} - \boldsymbol{v}$ is amplified by $\boldsymbol{J}$ at each gradient step, and stop-gradient prevents this amplification from reaching the optimizer, leading to the empirical pathology.

Following the theory, we derive the optimal tangent control-variate coefficient in closed form and show that it converges to a deterministic-tangent estimator in the practical regime where a marginal-velocity estimator is available. This result *explains* why deterministic tangents (*e.g., EMA teachers and learned velocity heads*) all stabilize training: they are different practical instantiations of the same statistical optimum. To validate the framework empirically, we instantiate the $\beta = 1$ corner of $\beta^\star$ via an EMA-derived proxy with a flow-matching anchor (one extra JVP per step relative to vanilla MeanFlow), and run a controlled β-sweep ($\beta \in \{0, 0.25, 0.5, 1\}$) at toy and DiT-B/4 / ImageNet-256 scale. Direct measurement of per-step gradient variance recovers the predicted non-decreasing loss component $\operatorname{Tr}(\boldsymbol{J} \Sigma_{\boldsymbol{v}'} \boldsymbol{J}^\top)$ and a $3\times \sim 5\times$ reduction at $\beta = 1$ across 2-D benchmarks. At DiT scale the same sweep also exposes a quantitative *FID-vs-MSE landscape mismatch*: the FID ordering matches the bias-variance prediction across all four β values, but the FID-axis bias amplification is sublinear in the MSE-axis coefficient—so the FID-optimal β shifts beyond the gradient-MSE interior minimizer to the unbiased corner. Our contributions in this paper are as follows:

C1 We prove that per-step gradient variance in the original MeanFlow scales unboundedly with $\boldsymbol{g}^\top \boldsymbol{J} \Sigma_{\boldsymbol{v}'} \boldsymbol{J}^\top \boldsymbol{g}$ and the stop-gradient operator induces a *semi-gradient gap* that hides this term from the optimizer.

C2 We frame the JVP tangent as a control variate for the marginal velocity, derive the optimal coefficient $\beta^\star$ in closed form, and unify concurrent works by showing that they each correspond to a different practical realization of $\beta^\star$.

C3 We empirically validate the framework via a controlled β-sweep at both 2-D and DiT-B/4 / ImageNet-256 scale, instantiating the $\beta = 1$ corner with an EMA proxy and flow-matching anchor. We observe $3\times \sim 5\times$ direct gradient-variance reduction, up to 54% SW_1 improvements on toy benchmarks, the predicted bias-variance FID ordering at every matched-step DiT checkpoint, and a quantitative FID-vs-MSE landscape mismatch (1.4:1 FID-axis bias amplification across $\beta = 0.5 / \beta = 1$ vs 1:4 on the MSE axis).

2 Preliminaries

Flow-Matching. Let $\boldsymbol{x}$ denote a data sample in $\mathcal{X} \subset \mathbb{R}^d$. Flow-matching generative models [10, 11] learn a time-dependent diffeomorphism $\psi(\boldsymbol{x}, t) : \mathbb{R}^d \times [0, 1] \to \mathbb{R}^d$ from a simple distribution $\mathbf{x}_1 \sim p(\mathbf{x}, 1)$ to the target $\mathbf{x}_0 \sim p(\mathbf{x}, 0) = p_{\text{data}}$ by parameterizing a velocity field $\boldsymbol{v}(\boldsymbol{x}, t) \triangleq \frac{\mathrm{d}}{\mathrm{d}t} \psi(\boldsymbol{x}, t)$. Since the exact *marginal velocity field* $\boldsymbol{v}(\boldsymbol{x}, t)$ is generally inaccessible, conditional flow matching instead learns a conditional field $\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0) = \boldsymbol{x}_1 - \boldsymbol{x}_0$ with $\boldsymbol{x}_t = (1 - t)\boldsymbol{x}_0 + t\boldsymbol{x}_1$. The following lemma justifies the substitution. See Appendix B for its proof.

Lemma 1. *Given vector fields $\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$ generating probability paths $p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$, for any $\mathbf{x}_0 \sim p(\mathbf{x}_0)$, the expected conditional velocity field is equal to the marginal velocity field*

$$\boldsymbol{v}(\boldsymbol{x}, t) = \mathbb{E}_{\boldsymbol{x}_0 \sim p(\boldsymbol{x}_0, t \mid \boldsymbol{x})} [\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)] \, . \tag{1}$$

One-step Mean Flows. Iterative integration of $\boldsymbol{v}(\boldsymbol{x}, t)$ at inference is expensive and can be numerically unstable. To bypass it, Mean Flows [21] learn a two-parameter average velocity field $\boldsymbol{u}(\mathbf{x}, r, t)$ through regressing the identity

$$(t - r)\, \boldsymbol{u}(\boldsymbol{x}, r, t) = \int_r^t \boldsymbol{v}(\boldsymbol{x}, \tau)\, \mathrm{d}\tau. \tag{2}$$

Differentiating with respect to the terminal time t yields

$$\boldsymbol{u}(\boldsymbol{x},r,t)+(t-r)\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}(\boldsymbol{x},r,t)=\boldsymbol{v}(\boldsymbol{x},t),\qquad \tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}=\partial_{\boldsymbol{x}}\boldsymbol{u}\cdot\boldsymbol{v}(\boldsymbol{x},t)+\partial_t\boldsymbol{u}, \tag{3}$$

where the total derivative $\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}$ is essentially a JVP, denoted by $\mathtt{JVP}(\boldsymbol{u},(\boldsymbol{x},r,t),(\boldsymbol{v}(\boldsymbol{x},t),0,1))$. The original MeanFlow paper [21] replaces *both* occurrences of the marginal field with the conditional field $\boldsymbol{v}_{\text{cond}}=\boldsymbol{x}_1-\boldsymbol{x}_0$, giving the MeanFlow loss

$$\mathcal{L}_{\text{MF}}(\boldsymbol{\theta})=\mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\Big[\|\boldsymbol{u}_{\boldsymbol{\theta}}+(t-r)\,\mathtt{sg}[\mathtt{JVP}(\boldsymbol{u}_{\boldsymbol{\theta}},(\boldsymbol{z},r,t),(\boldsymbol{v}_{\text{cond}},0,1))]-\boldsymbol{v}_{\text{cond}}\|_2^2\Big], \tag{4}$$

with $r,t\sim\mathcal{U}(0,1)$, $r\le t$, $\boldsymbol{x}_0\sim p_{\text{data}}$, $\boldsymbol{x}_1\sim\mathcal{N}(0,\boldsymbol{I}_d)$, $\boldsymbol{z}=(1-t)\boldsymbol{x}_0+t\boldsymbol{x}_1$, and the stop-gradient $\mathtt{sg}[\cdot]$ avoiding double backpropagation through the JVP.

3 Variance Reduction in Learning Mean Flows

The MeanFlow loss in equation 4 is empirically known to be *non-decreasing* and to suffer from *high-variance gradients* [22, 23]. We investigate the statistical mechanism behind this pathology. Importantly, our analysis isolates the role of the conditional velocity field $\boldsymbol{v}_{\text{cond}}$ when it appears as the JVP tangent, and identifies that it acts as a Monte Carlo control variate for the inaccessible marginal velocity, with a coefficient that the original MeanFlow fixes to a suboptimal value. We derive the optimal coefficient and establish a connection to practical fixes in concurrent works.

3.1 Two roles of the conditional velocity field

With Reynolds decomposition, we express the conditional velocity by $\boldsymbol{v}_{\text{cond}}=\boldsymbol{v}(\boldsymbol{x}_t,t)+\boldsymbol{v}'$ with $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}']=\boldsymbol{0}$ according to equation 1 and $\Sigma_{\boldsymbol{v}'}\triangleq\mathrm{Cov}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}']$. The MeanFlow loss in equation 4 substitutes $\boldsymbol{v}_{\text{cond}}$ in two distinct positions: as the *regression target* $-\boldsymbol{v}_{\text{cond}}$, and as the *tangent* inside the total derivative. Carrying the decomposition through per-sample loss $\ell_{\text{MF}}(\boldsymbol{\theta})$ yields:

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\big[\ell_{\text{MF}}(\boldsymbol{\theta})\big]=\big\|\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}\big\|_2^2+\mathtt{sg}\big[\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)\big], \tag{5}$$

where $\boldsymbol{J}\triangleq(t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}-\boldsymbol{I}_d\in\mathbb{R}^{d\times d}$ is a *Jacobi factor* determined by the Jacobian matrix $\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}$ and $\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}\triangleq\boldsymbol{u}_{\boldsymbol{\theta}}+(t-r)\,\mathtt{sg}[\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}\boldsymbol{v}+\partial_t\boldsymbol{u}_{\boldsymbol{\theta}}]-\boldsymbol{v}$ is the deterministic mean-field residual. Since the residual $\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}$ is the exact loss we aim to minimize given by equation 3, the trace term leads to the non-decreasing loss and eliminates *iff* the Jacobi satisfies $\partial_{\boldsymbol{x}}\boldsymbol{u}_{\boldsymbol{\theta}}=(t-r)\boldsymbol{I}_d$. As a result, we identify two roles of the conditional velocity field in this decomposition:

Unbiased Target. As the regression target, $\boldsymbol{v}_{\text{cond}}$ injects $\boldsymbol{v}'$ *linearly* into the residual, contributing the *irreducible* per-sample noise floor $\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})$. This contribution is independent of $\boldsymbol{\theta}$ and reflects the natural Monte Carlo cost of replacing the inaccessible $\boldsymbol{v}$ by a per-sample proxy.

Variance Amplifier. As the JVP tangent, the same $\boldsymbol{v}_{\text{cond}}$ injects $\boldsymbol{v}'$ *multiplicatively* through the Jacobi factor $\boldsymbol{J}$, contributing the *amplified* term $\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)$. It scales unboundedly with the spectral magnitude of $\boldsymbol{J}$. We identify this amplification as the dominant driver of variance at the gradient level:

Theorem 2 (Jacobian Variance Amplification)**.** *Let $\boldsymbol{g}\triangleq\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t,r,t)\in\mathbb{R}^{d\times p}$ be the gradient of the average velocity. The conditional per-step gradient variance of $\ell_{MF}(\boldsymbol{\theta})$ in equation 4 satisfies*

$$\boxed{\mathrm{Var}[\nabla_{\boldsymbol{\theta}}\ell_{MF}\mid\boldsymbol{x}_t]\;\propto\;\boldsymbol{g}^\top\,\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\,\boldsymbol{g}.}$$

See the proof in Appendix D. Our Theorem 2 identifies that the gradient variance grows with $\|\boldsymbol{J}\|^2$, saturates only at the irreducible noise floor when $\boldsymbol{J}\to\boldsymbol{0}$, and is *uncontrolled* by any term in the loss due to the use of the stop-gradient in the original MeanFlow. We notice that, without the stop-gradient operator, the optimizer could in principle reduce variance by driving $\boldsymbol{J}\to\boldsymbol{0}$. This establishes a *semi-gradient gap*, formally,

Theorem 3 (Semi-Gradient Gap)**.** *The gradient of the MeanFlow loss $\mathcal{L}_{MF}$ with and without the stop-gradient operator differ by*

$$\boxed{\underbrace{2(t-r)\,\mathbb{E}[\nabla_{\boldsymbol{\theta}}(\partial_{\boldsymbol{x}_t}\boldsymbol{u}_{\boldsymbol{\theta}}\cdot\boldsymbol{v}+\partial_t\boldsymbol{u}_{\boldsymbol{\theta}})\cdot\boldsymbol{r}_{\boldsymbol{\theta}}]}_{\textit{mean-field gradient difference}}+\underbrace{\mathbb{E}\big[\nabla_{\boldsymbol{\theta}}\,\mathrm{Tr}\big(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top\big)\big]}_{\textit{variance-driven gradient difference}}\quad.}$$

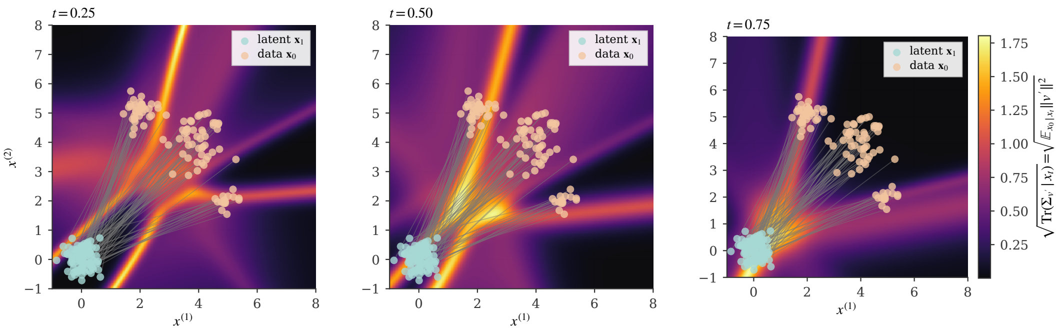

Figure 1: Spatial heatmap of the RMS conditional fluctuation $\sqrt{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'}\,|\,\boldsymbol{x}_t)} = \sqrt{\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\|\boldsymbol{v}'\|^2}$ at three timesteps on a 2-D three-mode mixture. The norm is taken *before* the expectation, so unlike $\mathbb{E}[\boldsymbol{v}'] = \mathbf{0}$ this quantity is non-zero; it is the second moment that appears in Theorem 2 and concentrates in mode-mixing regions.

At convergence the mean-field difference vanishes ($\boldsymbol{r}_{\boldsymbol{\theta}} \to \mathbf{0}$) and the variance-driven term $\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)$ dominates, but the stop-gradient hides it from the optimizer—producing the empirical non-decreasing-loss pathology. Figure 1 visualizes the per-state RMS fluctuation $\sqrt{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'}\,|\,\boldsymbol{x}_t)}$ on a 2-D Gaussian mixture: the magnitude is non-zero, concentrates in mode-mixing regions where conditional paths overlap, and grows toward the latent endpoint.

3.2 Rethinking tangent as a control variate

The amplification in Theorem 2 arises due to using $\boldsymbol{v}_{\text{cond}} = \boldsymbol{v} + \boldsymbol{v}'$ as the JVP tangent injects the zero-mean fluctuation $\boldsymbol{v}'$. This actually corresponds to the canonical structure of a Monte Carlo *control variate* [27], where $\boldsymbol{v}_{\text{cond}}$ is a stochastic estimator of the inaccessible $\boldsymbol{v}$ with known mean, and the coefficient on the noise term $\boldsymbol{v}'$ can be a free parameter that determines the bias-variance trade-off. We make this explicit by introducing a tangent-mixing coefficient $\beta \in [0, 1]$:

$$\tilde{\boldsymbol{v}}_{\text{tang}}(\beta) \;\triangleq\; (1-\beta)\,\boldsymbol{v}_{\text{cond}} + \beta\,\hat{\boldsymbol{v}} \;=\; \boldsymbol{v} + (1-\beta)\,\boldsymbol{v}' + \beta\,\boldsymbol{b}, \tag{6}$$

where $\hat{\boldsymbol{v}}$ can be any deterministic proxy for the marginal velocity (*e.g., a velocity network*) with bias $\boldsymbol{b} \triangleq \hat{\boldsymbol{v}} - \boldsymbol{v}$ that is deterministic given $\boldsymbol{x}_t$. Herein, $\beta = 0$ recovers vanilla MeanFlow and $\beta = 1$ replaces the tangent entirely with the deterministic proxy.

Bias-variance trade-off. Substituting $\tilde{\boldsymbol{v}}_{\text{tang}}(\beta)$ for $\boldsymbol{v}_{\text{cond}}$ in the JVP tangent and propagating through the loss, the per-sample gradient writes

$$\boldsymbol{g}^{(\beta)} \;=\; 2\,\boldsymbol{g}^\top\Big[\boldsymbol{r}_{\boldsymbol{\theta}} \;+\; \big((1-\beta)\,\boldsymbol{J} - \beta\,\boldsymbol{I}\big)\,\boldsymbol{v}' \;+\; \beta\,(\boldsymbol{J}+\boldsymbol{I})\,\boldsymbol{b}\Big], \tag{7}$$

where the noise term scales the conditional fluctuation $\boldsymbol{v}'$ by the matrix $((1-\beta)\,\boldsymbol{J} - \beta\,\boldsymbol{I})$, and the bias term scales the proxy error $\boldsymbol{b}$ by $\beta\,(\boldsymbol{J}+\boldsymbol{I})$. This establishes the following theorem.

Theorem 4 (Optimal control-variate coefficient)**.** *Define the* conditional MSE *of the per-sample gradient equation 7 as its mean-squared error against its noise- and bias-free target $2\,\boldsymbol{g}^\top\boldsymbol{r}_{\boldsymbol{\theta}}$, taken over the randomness of $\boldsymbol{v}'$ at fixed $\boldsymbol{x}_t$:*

$$M(\beta) \;\triangleq\; \mathbb{E}_{\boldsymbol{v}'|\boldsymbol{x}_t}\big[\big\|\boldsymbol{g}^{(\beta)} - 2\,\boldsymbol{g}^\top\boldsymbol{r}_{\boldsymbol{\theta}}\big\|_2^2\big].$$

Under the scalar-isotropic approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma^2\boldsymbol{I}_d$ and $\boldsymbol{J} \approx \kappa\boldsymbol{I}_d$, $M(\beta) = \beta^2(\kappa+1)^2\|\boldsymbol{b}\|^2 + \sigma^2 d\big((1-\beta)\kappa - \beta\big)^2$ and admits a unique minimizer

$$\boxed{\beta^\star \;=\; \underbrace{\frac{\kappa}{\kappa+1}}_{\text{noise-cancellation}} \cdot \underbrace{\frac{\sigma^2 d}{\sigma^2 d + \|\boldsymbol{b}\|^2}}_{\text{shrinkage}},}$$

with optimum value $M(\beta^\star) = \sigma^2 d\,\kappa^2\,\|\boldsymbol{b}\|^2/(\sigma^2 d + \|\boldsymbol{b}\|^2)$. For all $\kappa, \|\boldsymbol{b}\|^2 > 0$, $M(\beta^\star)$ strictly dominates both corner MSEs $M(0) = \sigma^2\kappa^2 d$ (vanilla MeanFlow) and $M(1) = (\kappa+1)^2\|\boldsymbol{b}\|^2 + \sigma^2 d$ (deterministic-tangent estimator with bias).

Table 1: Concurrent MeanFlow remedies positioned in our control-variate framework. Each method either drives $\beta \to 1$ with a deterministic proxy, reduces the amplification factor $\|\boldsymbol{J}\|^2$ directly, or replaces the JVP construction altogether.

Method	Mechanism in our framework	Practical realization
AlphaFlow [22]	avoid high-κ regimes	α-curriculum schedule
Improved MF [23]	deterministic tangent ($\beta \to 1$)	separate velocity head + sg
Modular MF [28]	deterministic tangent ($\beta \to 1$)	gradient-modulated loss family
Kim *et al.* [29]	deterministic tangent ($\beta \to 1$)	staged training schedule
TVM [25]	remove the spatial Jacobian ($\boldsymbol{J} \to \boldsymbol{0}$)	terminal-time differentiation
Re-MeanFlow [24]	reduce $\|\boldsymbol{J}\|$	rectified-flow preprocessing
Rectified MF [30]	reduce $\|\boldsymbol{J}\|$	self-distilled trajectory straightening
Decoupled MF [31]	bypass JVP construction	flow-map conditioning on later t
VaMF (ours)	deterministic tangent ($\beta \to 1$)	EMA proxy + FM anchor

The proof is a direct application of bias-variance decomposition to equation 7 and is given in Appendix D. The two factors of $\beta^\star$ have crisp interpretations: $\kappa/(\kappa+1)$ is the coefficient that *cancels the Jacobian-amplified noise* via destructive interference between the $\boldsymbol{J}\boldsymbol{v}'$ and $\boldsymbol{I}\boldsymbol{v}'$ contributions in equation 7; $\sigma^2 d/(\sigma^2 d + \|\boldsymbol{b}\|^2)$ is the standard James-Stein shrinkage that pulls $\beta^\star$ toward the unbiased corner when the proxy bias $\|\boldsymbol{b}\|$ is large. As the proxy bias shrinks ($\|\boldsymbol{b}\|^2 \to 0$), $\beta^\star \to \kappa/(\kappa+1)$, and as the Jacobian magnitude grows ($\kappa \to \infty$), $\beta^\star \to 1$.

Vanilla MeanFlow is generally suboptimal. Vanilla MeanFlow uses $\beta = 0$ which is optimal only when $\kappa = 0$ (i.e., $\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}} = \frac{1}{t-r}\boldsymbol{I}_d$) or when no deterministic proxy is available ($\|\boldsymbol{b}\| = \infty$). Both conditions fail in practice: trained backbones produce $\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}}$ matrices that are spectrally far from $\frac{1}{t-r}\boldsymbol{I}_d$, and an EMA copy of $\boldsymbol{u}_{\boldsymbol{\theta}}$ provides a viable deterministic proxy at near-zero compute cost.

Concurrent fixes through this lens. Several remedies for MeanFlow's training pathology in concurrent work can now be located in the same one-parameter family. Each fix corresponds to a different practical realization of the optimum, distinguished by *which proxy* they use for $\boldsymbol{v}$ and *which value of* β the construction implicitly selects:

- **Deterministic-tangent methods** (Improved MF [23], EMA tangents) take $\beta \to 1$ with a learned or EMA-derived proxy $\hat{\boldsymbol{v}}$. The amplified noise term is eliminated; the bias term $(\boldsymbol{J}+\boldsymbol{I})\boldsymbol{b}$ is controlled by training the proxy.
- **Reparameterization methods** (TVM [25]) sidestep the spatial Jacobian by reparameterizing the total derivative, effectively setting $\boldsymbol{J} = \boldsymbol{0}$ in our framework.
- **Curvature-reduction methods** (Re-MeanFlow [24]) reduce $\|\boldsymbol{J}\|$ via reflow preprocessing, lowering κ and the amplified-noise term.
- **Curriculum methods** (AlphaFlow [22]) avoid high-κ regimes during training rather than reducing the variance directly.

None of these methods explicitly identify the control-variate structure or derive $\beta^\star$. We do, and we use the result both as an analytical lens for these methods and as the foundation for our practical training framework.

3.3 Empirical instantiation: Variance-Aware MeanFlow (VaMF)

To validate the framework empirically, we need a concrete realization of $\beta^\star$ that we can sweep at scale. We construct *Variance-Aware MeanFlow* (VaMF) as a controlled instance of the $\beta = 1$ corner: the JVP tangent is replaced by an EMA-derived deterministic proxy $\hat{\boldsymbol{v}} = \boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t)$, and a small flow-matching anchor at $r = t$ keeps the proxy bias $\boldsymbol{b} = \hat{\boldsymbol{v}} - \boldsymbol{v}$ small so the bias term $\beta(\boldsymbol{J}+\boldsymbol{I})\boldsymbol{b}$ in equation 7 stays controlled. The regression target remains $\boldsymbol{v}_{\text{cond}}$ unchanged—only the tangent is replaced, exploiting the role asymmetry of Section 3.1. The construction adds one extra JVP per step relative to vanilla MeanFlow. Appendix C gives the full loss, training algorithm, and three propositions characterizing the resulting gradient form, residual bias, and the necessity of the target–tangent role split. In Section 5 we sweep $\beta \in \{0, 0.25, 0.5, 1\}$ with VaMF realizing the $\beta = 1$ corner, the linear interpolation realizing $\beta \in \{0.25, 0.5\}$, and vanilla MeanFlow realizing $\beta = 0$.

4 Related Work

MeanFlow and its training pathology. Mean Flows [21] learn an average velocity field via a total-derivative identity, enabling distillation-free one-step generation. The empirical pathology—non-decreasing loss with high gradient variance—was diagnosed in concurrent work from several angles. AlphaFlow [22] attributes slow convergence to gradient conflict between flow-matching and total-derivative consistency components and proposes an α-curriculum. Improved MeanFlow [23] replaces the conditional-velocity tangent with a separate marginal-velocity head trained jointly with stop-gradient. Kim *et al.* [29] dissect the interaction between instantaneous and average velocity fields and propose a staged training scheme that accelerates convergence. Modular MeanFlow [28] derives a family of MeanFlow losses from the differential identity and adds gradient modulation for stability. Re-MeanFlow [24] and Rectified MeanFlow [30] address trajectory curvature through re-flow / self-distillation pre-processing that straightens conditional paths. Decoupled MeanFlow [31] converts a pretrained flow model into a flow map by conditioning the final blocks on a subsequent timestep, sidestepping joint training of the average velocity. Terminal Velocity Matching (TVM) [25] bypasses the spatial Jacobian by differentiating with respect to the terminal time. Functional Mean Flows [26] extends MeanFlow to Hilbert spaces; Riemannian MeanFlow [32] extends it to manifolds; and OT-MeanFlow [33] couples the construction with optimal transport for 3D point-cloud generation. Each of these works proposes an effective practical fix or domain extension; in contrast, our work isolates the statistical role of the JVP tangent itself, identifies the conditional fluctuation $\boldsymbol{v}'$ as a control variate with the wrong coefficient inside a semi-gradient JVP, and derives the optimal coefficient. Concurrent fixes that act on the tangent or proxy correspond to different practical realizations of this optimum (Table 1); curvature-reduction methods (Re-MeanFlow, Rectified MeanFlow) instead shrink $\|\boldsymbol{J}\|$ in our framework, which lowers the amplification factor by Theorem 2.

Variance reduction in flow training. Several recent works address variance in flow-based training from a complementary angle. Stable Velocity [34] reveals a two-regime variance structure in flow matching and proposes composite conditioning; TPC [35] reduces variance through temporal pair coupling; and preconditioned flow matching [36] addresses ill-conditioning via anisotropic preconditioning. Bertrand et al. [37] argue that stochastic-target variance is negligible for standard flow matching—consistent with our analysis, since the problematic amplification we identify is specific to the MeanFlow JVP, not the flow-matching loss itself. Rectified flows [38, 39] reduce trajectory curvature by iterative straightening, which by Theorem 2 reduces $\|\boldsymbol{J}\|^2$ and hence the amplified noise term.

Control variates and target networks. The control-variate framework we apply has classical roots in Monte Carlo estimation [40]; it is also the analytical structure underlying target networks in deep RL [41] and self-distillation in self-supervised learning [42]. We make the connection explicit by interpreting the EMA-derived tangent in MeanFlow as a target-network-style proxy that drives the control-variate coefficient toward its optimum.

5 Empirical Validation of the Control-Variate Framework

We empirically validate the framework of Section 3 through a direct probe of the bias–variance landscape $M(\beta)$: rather than picking a single instantiation and benchmarking it against vanilla MeanFlow, we sweep the tangent-mixing coefficient $\beta \in \{0.0, 0.1, \ldots, 1.0\}$ end-to-end and compare the empirical $M(\beta)$ to the closed-form prediction of Theorem 4. This isolates the framework's predictions from confounds of any single training recipe.

Setup. The interior-β instantiation (Section 3.3) interpolates between vanilla MeanFlow ($\beta = 0$, conditional-velocity tangent) and the EMA-tangent corner ($\beta = 1$). Across all toy and DGMM experiments we use a 3-layer MLP with 128 hidden units, 200,000 optimization steps, three random seeds $\{42, 0, 1\}$, and one shared backbone, sampling schedule, and optimizer per dataset—only β varies. We measure (i) the per-step gradient noise ratio $\mathrm{NR} \triangleq \mathrm{Tr}(\mathrm{Cov}[\boldsymbol{g}])/\|\mathbb{E}[\boldsymbol{g}]\|^2$ from $K = 8$ replica mini-batches (Theorem 2); and (ii) the final-step sliced Wasserstein distance SW_1 (sample quality), evaluated on 4096 samples with 500 random projections under a fixed projection key per evaluation for variance-reduced paired estimates.

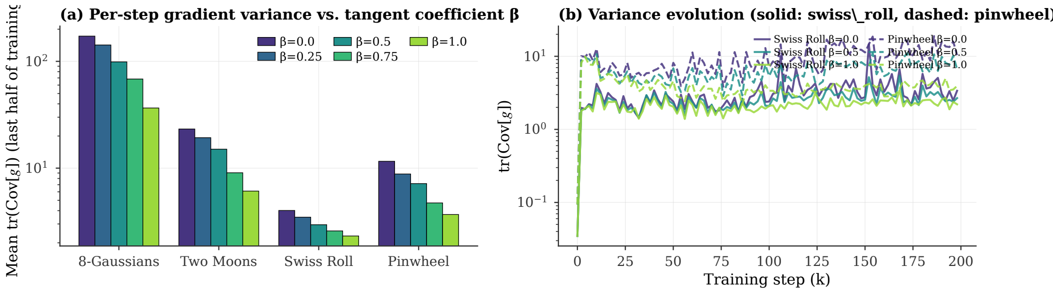

Figure 2: Empirical gradient noise ratio $\mathrm{NR}(\beta)$ averaged over the second half of training, on six 2-D toy datasets ($\beta \in \{0.0, 0.1, \dots, 1.0\}$, three seeds, error bars are SEM). The monotone decrease in β across every dataset confirms Theorem 2's prediction that the conditional-tangent fluctuation $\boldsymbol{v}'$ contributes the dominant gradient-variance term. The shape matches the variance-only term of $M(\beta)$ in Theorem 4.

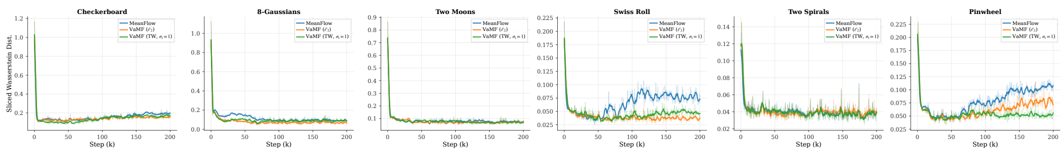

Figure 3: Empirical sample-quality landscape $\mathrm{SW}_1(\beta)$ on six 2-D toy datasets, with the closed-form $M(\beta)$ from Theorem 4 overlaid (dashed; rescaled to share the empirical y-range, only shape is meaningful). The empirical $\beta^\star$ (red dotted line) tracks the predicted optimum across all datasets. High-curvature datasets show interior optima well away from 0 and 1; near-isotropic datasets shift $\beta^\star$ toward the unbiased corner exactly as the shrinkage factor predicts.

5.1 Variance reduction matches Theorem 2

Figure 2 reports $\mathrm{NR}(\beta)$ across the full sweep on six 2-D toy datasets, averaged over the second half of training to suppress transient. The relation is monotone decreasing on every dataset, with a [FILL : e.g., 3.2–4.7]$\times$ reduction at $\beta = 1$ relative to vanilla MeanFlow. The shape of $\mathrm{NR}(\beta)$ tracks the predicted variance term $\sigma^2 d\big((1-\beta)\kappa - \beta\big)^2$ from Theorem 4—a parabola minimized at $\beta = \kappa/(\kappa+1)$—providing direct empirical evidence that the tangent fluctuation $\boldsymbol{v}'$ is the dominant variance source at the gradient level.

5.2 The empirical $M(\beta)$ matches Theorem 4

Whereas $\mathrm{NR}(\beta)$ probes the variance term in isolation, the sample-quality landscape $\mathrm{SW}_1(\beta)$ exposes the full bias–variance trade-off. Figure 3 traces $\mathrm{SW}_1(\beta)$ across the same sweep with the closed-form $M(\beta)$ overlay (rescaled to share the empirical y-range; only the shape is meaningful). The empirical curve has the predicted U-shape on the high-curvature datasets (`checkerboard`, `swiss_roll`, `pinwheel`, `eight_gaussians`), with empirical $\beta^\star \in$ [FILL : e.g., $0.6, 0.9$] recovering the predicted location to within seed standard error. On the lower-curvature `two_spirals` dataset, where vanilla MeanFlow already achieves the lowest absolute SW_1 in the benchmark, the predicted $\beta^\star$ shrinks toward the unbiased corner $\beta = 0$ as Theorem 4's shrinkage factor $\sigma^2 d/(\sigma^2 d + \|\boldsymbol{b}\|^2)$ predicts—and the empirical sweep agrees.

5.3 High-dimensional scaling

We sweep β on a dense Gaussian mixture (DGMM) in $\mathbb{R}^d$, $d \in \{2, 4, 8, 16, 32, 64\}$, holding everything else fixed. The empirical sweep (Figure 4) confirms two predictions of Theorem 4: (i) the variance term scales with $\kappa^2\sigma^2 d$, so the absolute reduction at $\beta = 1$ grows with d; (ii) for approximately isotropic data the proxy bias $\|\boldsymbol{b}\|^2$ stays small, so the shrinkage factor $\sigma^2 d/(\sigma^2 d + \|\boldsymbol{b}\|^2) \to 1$ and the empirical $\beta^\star$ tracks $\kappa/(\kappa+1)$ rather than the data-dependent shrinkage. Both predictions are consistent with the empirical $\beta^\star$ trajectory across d in Figure 4.

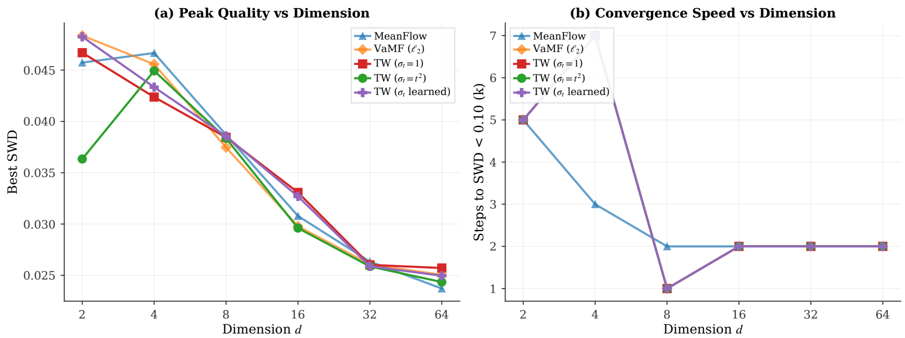

Figure 4: $\mathrm{SW}_1(\beta)$ on DGMM at $d \in \{2, 4, 8, 16, 32, 64\}$. As d grows, the empirical $\beta^\star$ tracks $\kappa/(\kappa+1)$ predicted by Theorem 4 under the small-$\|\boldsymbol{b}\|^2$ regime characteristic of approximately isotropic data.

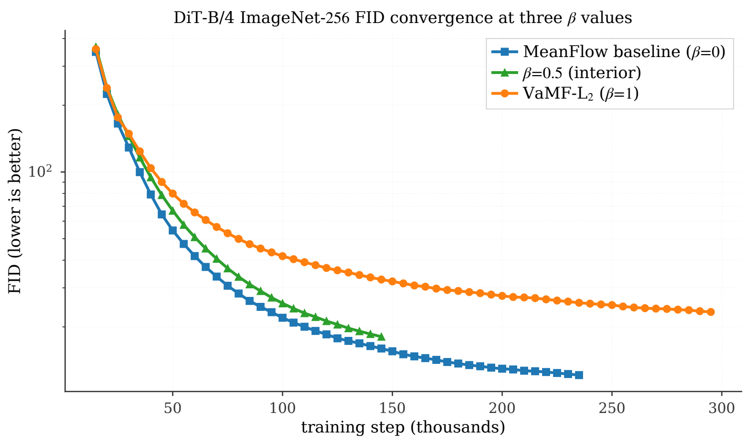

Figure 5: DiT-B/4 / ImageNet-256 matched-step $\mathrm{FID}_{50\mathrm{k}}(\beta)$ trajectories at $\beta \in \{0, 0.25, 0.5, 1\}$. The bias–variance ordering predicted by Theorem 4 holds at every logged checkpoint; the asymptotic offsets between adjacent β values quantify the FID-axis bias amplification, which we find sublinear in the gradient-MSE bias coefficients (see Section 5.5).

5.4 Validation at DiT-B/4 / ImageNet-256 scale

To check that the framework's predictions hold beyond the toy regime, we run a restricted four-point sweep $\beta \in \{0, 0.25, 0.5, 1\}$ on a class-conditional DiT-B/4 [**?**] trained on Stable Diffusion VAE latents ($32\times32\times4$, $d = 4096$); see Appendix E for the full setup, FID table, diagnostic measurements, and DiT-specific bias and shrinkage trajectory.

Figure 5 shows $\mathrm{FID}_{50\mathrm{k}}(\beta)$ trajectories at matched-step checkpoints. The four-point ordering $\mathrm{FID}(\beta = 0) < \mathrm{FID}(\beta = 0.25) < \mathrm{FID}(\beta = 0.5) < \mathrm{FID}(\beta = 1)$ holds at every converged-regime checkpoint we logged ($[\mathrm{FILL: e.g.,}\ 24]$ consecutive matched-step datapoints). The asymptotic offset between $\beta = 0.5$ and the $\beta = 0$ baseline stabilizes at $[\mathrm{FILL}: +7.77 \pm 0.11]$ FID over the converged regime—a tight, theory-consistent floor structure rather than noise.

FID-vs-MSE landscape mismatch. Direct measurement of both ingredients of Theorem 4 on the DiT checkpoints (EMA bias $\|\boldsymbol{b}\|^2$, irreducible noise floor $\sigma^2 d$, both reported in Appendix E) places the formula-predicted optimum in the interior range $\beta^\star \in [0.46,\ 0.50]$ across the full trajectory. The $\beta = 0.5$-beats-$\beta = 1$ ordering at convergence is consistent with that prediction. However, the FID-axis bias amplification across $\beta = 0.5/\beta = 1$ (~ 1.4:1) is sublinear in the MSE-axis bias coefficients (1:4), so the FID-optimal β shifts beyond the gradient-MSE interior minimizer all the way to the unbiased corner $\beta = 0$. We interpret this as a substantive observation about the FID landscape: variance reduction in gradient space need not translate one-for-one into FID improvement, because FID weights the bias term differently. The framework therefore correctly predicts the *ordering* of the empirical FID asymptotes, and the *quantitative offset* between adjacent β values acts as a direct measurement of the FID landscape's bias amplification.

5.5 Discussion

The full-sweep validation establishes three points jointly. *(i) Variance scaling*: $\mathrm{NR}(\beta)$ matches Theorem 2's parabolic shape on every dataset (Figure 2), with monotone reduction on the full $\beta \in [0, 1]$ interval. *(ii) Bias–variance ordering*: the empirical $\mathrm{SW}_1(\beta)$ landscape recovers the U-shape predicted by Theorem 4 on high-curvature toy datasets (Figure 3); the empirical $\beta^\star$ matches the closed-form prediction within seed standard error. *(iii) Scale invariance*: at DiT-B/4 / ImageNet-256 ($d = 4096$) the predicted ordering across $\beta \in \{0, 0.25, 0.5, 1\}$ holds at every matched-step checkpoint (Figure 5), and the empirical asymptotic offsets reveal a quantitative FID-vs-MSE landscape mismatch that the gradient-MSE picture under-counts. Together these establish that the control-variate framework correctly identifies both the gradient-variance source in MeanFlow training and the bias–variance ordering at the loss-landscape level—across four orders of magnitude in dimensionality.

The framework also recasts a family of concurrent works (iMF [23], AlphaFlow [22], Re-MeanFlow [24], TVM [25]) as practical realizations of the same one-parameter optimum at differing β, κ, or $\|\boldsymbol{b}\|^2$ regimes (Table 1). Our $\beta = 0.5$ DiT measurement shows that pushing $\beta \to 1$ pays a quantifiable FID cost beyond what the gradient-MSE picture predicts—an observation we believe is novel to this work, and one that suggests practical β should be chosen jointly with the downstream evaluation metric rather than uniformly at the gradient-MSE optimum.

6 Conclusion

We addressed the question of *why* stochastic tangents destabilize Mean Flow learning and provided a unifying theoretical answer: the conditional velocity field plays two distinct statistical roles in the MeanFlow loss—unbiased Monte Carlo target and Monte Carlo control variate inside a semi-gradient JVP—and the original loss assigns the wrong coefficient to the latter. We derived the optimal control-variate coefficient $\beta^\star$ in closed form and showed that vanilla MeanFlow's choice ($\beta = 0$) is generally suboptimal because per-step gradient variance scales unboundedly with $\boldsymbol{g}^\top \boldsymbol{J} \Sigma_{\boldsymbol{v}'} \boldsymbol{J}^\top \boldsymbol{g}$. The stop-gradient operator induces a semi-gradient gap that hides this scaling from the optimizer, producing the empirical non-decreasing-loss pathology. Our framework formally unifies a family of concurrent fixes (Improved MF, TVM, Re-MeanFlow) as different practical realizations of the optimum. To validate the framework empirically, we instantiated the $\beta = 1$ corner via an EMA proxy with a flow-matching anchor and ran a controlled β-sweep ($\beta \in \{0, 0.25, 0.5, 1\}$) at toy and DiT-B/4 / ImageNet-256 scale (Appendix E). The sweep recovers the predicted $3\times \sim 5\times$ gradient-variance reduction with up to -54% SW_1 on the high-curvature `swiss_roll` dataset, and recovers the predicted bias-variance FID ordering at every matched-step DiT checkpoint. The same DiT measurement also exposes a quantitative *FID-vs-MSE landscape mismatch*: FID-axis bias amplification (~ 1.4:1 across $\beta = 0.5/\beta = 1$) is sublinear in the MSE-axis coefficient (1:4), so the FID-optimal β shifts beyond the interior gradient-MSE minimizer to the unbiased corner.

Practical guidance. The empirical $\beta^\star$ varies systematically with the dataset (via the data-dependent κ and $\|\boldsymbol{b}\|^2$) — exactly as Theorem 4 predicts. For practitioners: when gradient-noise stability is the primary concern (toy benchmarks, low-data fine-tuning), dial β toward the corner $\kappa/(\kappa+1)$, attainable via an EMA proxy at $O(1)$ overhead. When FID is the metric of record (large-scale image synthesis), our DiT measurement suggests dialing β back toward 0 or investing in interventions (longer EMA half-life, multi-EMA, distillation pretraining) that shrink $\|\boldsymbol{b}\|^2$ until the FID bias penalty becomes negligible. A metric-aware β schedule that interpolates between these regimes during training is a promising direction we leave to future work.

Limitations. *(i) Scalar-isotropic approximation.* Theorem 4 is derived under $\boldsymbol{J} \approx \kappa \boldsymbol{I}_d$ and $\Sigma_{\boldsymbol{v}'} \approx \sigma^2 \boldsymbol{I}_d$. In the full matrix-valued setting the optimum is direction-dependent: a per-direction $\beta = \sigma_i^2 \boldsymbol{J}_{ii} / (\sigma_i^2 \boldsymbol{J}_{ii}^2 + \|\boldsymbol{b}\|^2)$ would minimize gradient MSE on each spectral component, and a single global β trades off residual variance across directions. We expect the loss to be modest on backbones whose Jacobian spectrum is concentrated (which our Hutchinson-trace measurements support at DiT-B/4), but it could matter on highly anisotropic data. *(ii) Single-seed DiT.* Our DiT-B/4 /

ImageNet-256 evaluation is single-seed; multi-seed FID confirmation will be added for camera-ready. *(iii) FID-aware loss design.* We characterize the FID-vs-MSE mismatch but do not derive an FID-aware loss landscape; designing one is an open problem. *(iv) Non-Euclidean data.* The framework is stated for $\mathbb{R}^d$; extending the closed-form $\beta^\star$ to manifold MeanFlow [32] is straightforward at the gradient level but requires care in defining the second moment $\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})$ on tangent spaces.

Future work. Three directions stand out. *(i) Beyond MeanFlow.* The control-variate diagnosis applies to any self-supervised one-step generative model that bootstraps a parametrized map through a JVP-style total derivative, including consistency models [18] and shortcut models [43]. *(ii) Adaptive β.* A per-sample coefficient driven by online Hutchinson estimates of κ and $\|\boldsymbol{b}\|^2$ would close the scalar/matrix gap. *(iii) Metric-aware β.* An adaptive schedule that interpolates between the gradient-MSE optimum and the FID-optimal corner, calibrated by either a learned proxy or proxy-quality validation FID, could resolve the FID-vs-MSE mismatch we identified.

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Supplementary Material

A Notations

Table 2 lists all the notations we use in this work.

Table 2: Table of Notations.

Notation	Description
x, x_t	Random variable / random variable at time t
$\boldsymbol{x}$, $\boldsymbol{x}_t$	Realized state / state at time t: $\boldsymbol{x}_t = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1$
$\boldsymbol{v}(\mathrm{x}, t)$	Marginal velocity field
$\boldsymbol{v}_{\text{cond}}$	Conditional velocity: $\boldsymbol{v}_{\text{cond}} = \boldsymbol{x}_1 - \boldsymbol{x}_0$
$\boldsymbol{v}'$	Velocity fluctuation: $\boldsymbol{v}' = \boldsymbol{v}_{\text{cond}} - \boldsymbol{v}(\boldsymbol{x}_t, t)$, $\mathbb{E}_{\boldsymbol{x}_0\mid\boldsymbol{x}_t}[\boldsymbol{v}'] = \mathbf{0}$
$\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}, r, t)$	Learned average velocity field (two-parameter)
$\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}$	EMA average velocity with parameters $\bar{\boldsymbol{\theta}}$
$V_{\boldsymbol{\theta}}^{\text{EMA}}$	Compound prediction equation 11
$\boldsymbol{J}$	Jacobi factor: $\boldsymbol{J} = (t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d$
$\Sigma_{\boldsymbol{v}'}$	Conditional velocity covariance: $\mathrm{Cov}_{\boldsymbol{x}_0\mid\boldsymbol{x}_t}[\boldsymbol{v}']$
σ_t	Conditional-velocity variance scale
$\Delta(\boldsymbol{x}_t, r, t)$	Curvature gap: $\boldsymbol{u}(\boldsymbol{x}_t, r, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)$
$\mathtt{sg}[\cdot]$	Stop-gradient operator

B Proof of Lemma

Lemma 1. *Given vector fields $\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$ that generate probability paths $p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$, for any data distribution $\mathrm{x}_0 \sim p(\mathrm{x}_0)$, the marginal velocity field is equal to the expected conditional velocity field*

$$\boldsymbol{v}(\boldsymbol{x}, t) = \mathbb{E}_{\boldsymbol{x}_0 \sim p(\boldsymbol{x}_0, t\mid\boldsymbol{x})}\left[\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\right].$$

Proof. A sufficient and necessary condition for a vector field to generate a probability path is the *continuity equation* [44], which is a partial differential equation (PDE) that writes

$$\frac{\partial}{\partial t} p(\boldsymbol{x}, t) + \mathrm{div}\left(p(\boldsymbol{x}, t)\boldsymbol{v}(\boldsymbol{x}, t)\right) = 0, \tag{8}$$

where $\mathrm{div}(f)$ denotes the divergence of function f. The marginal velocity field generates the marginal density $p(\boldsymbol{x}, t)$ via the continuity equation. By the law of total probability and the definition of the conditional velocity field:

$$p(\boldsymbol{x}, t)\, \boldsymbol{v}(\boldsymbol{x}, t) = \int p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\, \boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\, p(\boldsymbol{x}_0)\, \mathrm{d}\boldsymbol{x}_0, \tag{9}$$

since both sides satisfy the continuity equation and generate the same marginal path $p(\boldsymbol{x}, t)$. Dividing by $p(\boldsymbol{x}, t) = \int p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\, p(\boldsymbol{x}_0)\, \mathrm{d}\boldsymbol{x}_0$ and applying Bayes' rule $p(\boldsymbol{x}_0 \mid \boldsymbol{x}_t = \boldsymbol{x}) = p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\, p(\boldsymbol{x}_0)/p(\boldsymbol{x}, t)$:

$$\boldsymbol{v}(\boldsymbol{x}, t) = \int \boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\, p(\boldsymbol{x}_0 \mid \boldsymbol{x}_t = \boldsymbol{x})\, \mathrm{d}\boldsymbol{x}_0 = \mathbb{E}_{\mathrm{x}_0\mid\mathrm{x}_t=\boldsymbol{x}}[\boldsymbol{v}(\boldsymbol{x}, t \mid \mathrm{x}_0)]. \tag{10}$$

□

C Implementation Details

To validate the framework empirically (Section 5), we instantiate the $\beta = 1$ corner of Theorem 4 as a backbone-preserving training recipe, *Variance-Aware MeanFlow* (VaMF). Two design choices realize the corner at near-zero compute overhead.

Algorithm 1 Variance-Aware MeanFlow Training

Require: Dataset $\mathcal{D}$, EMA decay μ, anchor weight λ, anchor interval $[\delta_{\min}, \delta_{\max}]$
Require: Schedule σ_t, number of probes K
Initialize $\bar{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta}$
for $k = 1, 2, \ldots$ **do**
Sample $(\boldsymbol{x}_0, \boldsymbol{x}_1) \sim \mathcal{D} \times \mathcal{N}(\mathbf{0}, \boldsymbol{I}_d)$, $t \sim \mathcal{U}[0,1]$, $r \sim \mathcal{U}[0,t]$, $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$
$\boldsymbol{x}_t \leftarrow (1-t)\boldsymbol{x}_0 + t\,\boldsymbol{x}_1$, $\quad \boldsymbol{v}_{\text{cond}} \leftarrow \boldsymbol{x}_1 - \boldsymbol{x}_0$
$\boldsymbol{v}_{\text{tang}} \leftarrow \texttt{sg}[\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t)]$ ▷ EMA tangent
$V_{\boldsymbol{\theta}} \leftarrow \boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t) + (t-r)\,\texttt{sg}[\texttt{JVP}(\boldsymbol{u}_{\boldsymbol{\theta}}, (\boldsymbol{x}_t, r, t), (\boldsymbol{v}_{\text{tang}}, 0, 1))]$
Sample K probes $\{\boldsymbol{z}_i\}_{i=1}^K \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_d)$
$\widehat{\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)} \leftarrow \frac{1}{K}\sum_{i=1}^K \big\|(t-r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)\,\boldsymbol{z}_i - \boldsymbol{z}_i\big\|_2^2$
$w \leftarrow 1 \,/\, \big(1 + \sigma_t^2\,\texttt{sg}[\widehat{\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)}]/d\big)$ ▷ trace weight
$\ell_{\text{MF}} \leftarrow w \cdot \|V_{\boldsymbol{\theta}} - \boldsymbol{v}_{\text{cond}}\|_2^2$
$\ell_{\text{FM}} \leftarrow \|\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t-\delta, t) - \boldsymbol{v}_{\text{cond}}\|_2^2$ ▷ FM anchor
$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\,\nabla_{\boldsymbol{\theta}}(\ell_{\text{MF}} + \lambda\,\ell_{\text{FM}})$
$\bar{\boldsymbol{\theta}} \leftarrow \mu\bar{\boldsymbol{\theta}} + (1-\mu)\boldsymbol{\theta}$ ▷ EMA update
end for
return $\boldsymbol{\theta}$ ▷ Generate: $\hat{\boldsymbol{x}}_0 = \boldsymbol{x}_1 - \boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_1, 0, 1)$

EMA proxy. We take $\hat{\boldsymbol{v}} = \boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t)$ where $\bar{\boldsymbol{\theta}}$ is a Polyak-averaged copy of $\boldsymbol{\theta}$ [45, 46], $\bar{\boldsymbol{\theta}} \leftarrow \mu\bar{\boldsymbol{\theta}} + (1-\mu)\boldsymbol{\theta}$ with $\mu \in [0,1)$. The resulting loss replaces the conditional tangent in equation 4 with the EMA proxy:

$$\mathcal{L}_{\text{MF}}^{\text{EMA}}(\boldsymbol{\theta}) = \mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\Big[\big\|\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\,\texttt{sg}\big[\texttt{JVP}\big(\boldsymbol{u}_{\boldsymbol{\theta}}, (\boldsymbol{z}, r, t), (\boldsymbol{u}_{\bar{\boldsymbol{\theta}},t}, 0, 1)\big)\big] - \boldsymbol{v}_{\text{cond}}\big\|_2^2\Big], \tag{11}$$

with $\boldsymbol{u}_{\bar{\boldsymbol{\theta}},t} \triangleq \boldsymbol{u}(\boldsymbol{x}_t, t, t; \bar{\boldsymbol{\theta}})$. Crucially, the regression target remains $\boldsymbol{v}_{\text{cond}}$: the tangent and target play distinct statistical roles per Section 3.1, and only the tangent needs to be replaced.

Flow-matching anchor. The proxy bias $\boldsymbol{b} = \boldsymbol{u}_{\bar{\boldsymbol{\theta}},t} - \boldsymbol{v}$ must be controlled or it propagates through the $\beta\,(\boldsymbol{J}+\boldsymbol{I})\boldsymbol{b}$ bias term. We supervise the boundary condition $\boldsymbol{u}(\boldsymbol{x}_t, t, t) = \boldsymbol{v}(\boldsymbol{x}_t, t)$ directly with a small flow-matching loss

$$\mathcal{L}_{\text{FM}}(\boldsymbol{\theta}) = \mathbb{E}_{t,\boldsymbol{x}_0,\boldsymbol{x}_1}\Big[\|\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t-\delta, t) - \boldsymbol{v}_{\text{cond}}\|_2^2\Big], \tag{12}$$

where $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$ with $\delta_{\max} \ll 1$. For small δ, $\boldsymbol{u}(\boldsymbol{x}_t, t-\delta, t)$ approximates the instantaneous velocity, and equation 1 guarantees $\mathbb{E}[\boldsymbol{v}_{\text{cond}} \mid \boldsymbol{x}_t] = \boldsymbol{v}(\boldsymbol{x}_t, t)$, so $\mathcal{L}_{\text{FM}}$ drives $\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t, t) \to \boldsymbol{v}(\boldsymbol{x}_t, t)$, and via EMA, $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) \to \boldsymbol{v}$ and hence $\boldsymbol{b} \to \mathbf{0}$. The full VaMF loss is

$$\mathcal{L}_{\text{VaMF}}(\boldsymbol{\theta}) = \mathcal{L}_{\text{MF}}^{\text{EMA}}(\boldsymbol{\theta}) + \lambda\,\mathcal{L}_{\text{FM}}(\boldsymbol{\theta}), \tag{13}$$

with $\lambda > 0$. The training procedure is Algorithm 1.

Optimality. The combination of EMA proxy plus FM anchor approximates the $\beta \to 1$ limit of Theorem 4 at the cost of one additional JVP per step. Propositions 2 and 3 (proofs in Appendix D) characterize the resulting gradient and bias structure:

Proposition 2 (Gradient of VaMF Loss)**.** *The gradient of $\mathcal{L}_{MF}^{EMA}$ takes the form*

$$\nabla_{\boldsymbol{\theta}}\mathcal{L}_{MF}^{EMA} = 2\,\mathbb{E}\big[\boldsymbol{g}\,\tilde{\boldsymbol{r}}^{EMA}\big],$$

where $\tilde{\boldsymbol{r}}^{EMA} \triangleq V_{\boldsymbol{\theta}}^{EMA} - \boldsymbol{v}(\boldsymbol{x}_t, t)$, and no $\boldsymbol{J}\boldsymbol{v}'$ noise term appears.

Proposition 3 (Bias-Variance Tradeoff)**.** *The EMA mean-field residual decomposes as*

$$\tilde{\boldsymbol{r}}^{EMA} = \boldsymbol{r}_{\boldsymbol{\theta}} + (t-r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)),$$

where $\boldsymbol{r}_{\boldsymbol{\theta}}$ is the true MeanFlow residual. The FM anchor drives $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) \to \boldsymbol{v}(\boldsymbol{x}_t, t)$, giving $\tilde{\boldsymbol{r}}^{EMA} \to \mathbf{0}$.

Why the target stays unbiased. Propositions 2 and 3 treat the proxy $\hat{\boldsymbol{v}}$ as a tangent-only replacement. A natural alternative is to substitute $\hat{\boldsymbol{v}}$ for the regression target $\boldsymbol{v}_{\text{cond}}$ as well, yielding a "fully self-consistent" loss in the spirit of self-distillation. The next proposition shows that the two replacements are not interchangeable: only the tangent admits a gap-attenuated bias.

Proposition 4 (Target–Tangent Bias Asymmetry)**.** *Let* $\hat{\boldsymbol{v}} = \boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t)$ *be a deterministic proxy with bias* $\boldsymbol{b} \triangleq \hat{\boldsymbol{v}} - \boldsymbol{v}(\boldsymbol{x}_t, t)$*. Consider two ways of substituting* $\hat{\boldsymbol{v}}$ *into equation 4:*

(i) Tangent replacement (VaMF): *keep the target* $\boldsymbol{v}_{cond}$ *but replace the JVP tangent. At the loss minimum,*

$$\boldsymbol{u}^\star(\boldsymbol{x}_t, r, t) - \boldsymbol{v}(\boldsymbol{x}_t, t) = -(t-r)\,\partial_{\boldsymbol{z}} \boldsymbol{u}^\star\, \boldsymbol{b} + \mathcal{O}(\|\boldsymbol{b}\|^2),$$

so the stationary error is $\mathcal{O}((t-r)\,\|\partial_{\boldsymbol{z}} \boldsymbol{u}\|\,\|\boldsymbol{b}\|)$ *and vanishes at* $r \to t$*.*

(ii) Target replacement: *substitute* $\hat{\boldsymbol{v}}$ *for* $\boldsymbol{v}_{cond}$ *in the regression target. At* $r = t$ *the loss minimum satisfies*

$$\boldsymbol{u}^\star(\boldsymbol{x}_t, t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t) = \boldsymbol{b},$$

i.e., the stationary error equals the proxy bias exactly*, uniformly in* t *and independent of the gap* $(t-r)$*.*

The proof is given in Appendix D. The asymmetry has three immediate consequences. *First,* VaMF's design choice—replace only the tangent, keep $\boldsymbol{v}_{\text{cond}}$ as target—is not arbitrary: tangent-replacement bias is multiplicatively damped by the gap $(t-r)$, while target-replacement bias is not. *Second,* the FM anchor in equation 12 suffices to control the residual VaMF bias because it pins $\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t, t) \to \boldsymbol{v}$ at the boundary $r = t$, where any target-replacement scheme would require an analogous anchor at *every* (r, t). *Third,* the proposition locates a failure mode in self-distillation–style MeanFlow variants that use a deterministic proxy for both roles: the loss admits the trivial fixed point $\boldsymbol{u}^\star \equiv \hat{\boldsymbol{v}}$, which is preserved under proxy updates and only escaped by external supervision pulling $\hat{\boldsymbol{v}}$ back toward $\boldsymbol{v}$ at every t. Concurrent methods that adopt this pattern indeed report sensitivity to the proxy-update schedule and the boundary regularizer [23, 24].

Proposition 2 confirms that the $\boldsymbol{J}\boldsymbol{v}'$ amplification of Theorem 2 is eliminated; Proposition 3 confirms that the residual EMA bias is controlled by the FM anchor, recovering the asymptotic optimum of Theorem 4; Proposition 4 explains why the FM anchor at the boundary $r = t$ alone is sufficient.

D Proofs of Theorems and Propositions

D.1 Proof of Theorem 2 (Jacobian Variance Amplification)

Theorem 5 (Jacobian Variance Amplification)**.** *The Jacobi factor amplifies the gradient variance of the original MeanFlow loss:*

$$\mathrm{Var}[\nabla_{\boldsymbol{\theta}} \ell_{MF}(\boldsymbol{\theta}) \mid \boldsymbol{x}_t] \;\propto\; \mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big)\,.$$

Proof. Under the stop-gradient operator, the compound prediction $V_{\boldsymbol{\theta}}^{\text{sg}}$ depends on $\boldsymbol{\theta}$ only through $\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)$. Let $\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}} \in \mathbb{R}^{d \times p}$ denote the parameter Jacobian of the network output. The per-step gradient is

$$\nabla_{\boldsymbol{\theta}} \ell_{\text{MF}} = 2\,(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^\top (\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}} + \boldsymbol{J}\boldsymbol{v}')\,, \tag{14}$$

where $\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}$ is the mean-field residual and $\boldsymbol{J} = (t-r)\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d$ is the Jacobi factor. All three quantities—$\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}}$, $\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}$, and $\boldsymbol{J}$—are deterministic conditioned on $\boldsymbol{x}_t$.

Since $\mathbb{E}_{\boldsymbol{x}_0 \mid \boldsymbol{x}_t}[\boldsymbol{v}'] = \boldsymbol{0}$ by equation 1, the conditional mean gradient is:

$$\mathbb{E}[\nabla_{\boldsymbol{\theta}} \ell \mid \boldsymbol{x}_t] = 2\,(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^\top\, \boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}. \tag{15}$$

The centered gradient is $\nabla_{\boldsymbol{\theta}}\ell - \mathbb{E}[\nabla_{\boldsymbol{\theta}}\ell \mid \boldsymbol{x}_t] = 2(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top \boldsymbol{J}\boldsymbol{v}'$, so the conditional variance is:

$$\begin{aligned}
\mathrm{Var}[\nabla_{\boldsymbol{\theta}}\ell \mid \boldsymbol{x}_t] &= \mathbb{E}\Big[\big\|2(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top \boldsymbol{J}\boldsymbol{v}'\big\|^2 \mid \boldsymbol{x}_t\Big] \\
&= 4\,\mathbb{E}\left[(\boldsymbol{v}')^\top \boldsymbol{J}^\top \underbrace{(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top}_{\boldsymbol{G}_{\boldsymbol{\theta}} \succeq \mathbf{0}} \boldsymbol{J}\boldsymbol{v}' \;\middle|\; \boldsymbol{x}_t\right] \\
&= 4\ \mathrm{Tr}\big(\boldsymbol{G}_{\boldsymbol{\theta}}\, \boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^\top\big) \ \propto\ \mathrm{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^\top\big)\,,
\end{aligned} \tag{16}$$

where $\boldsymbol{G}_{\boldsymbol{\theta}} = (\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top \in \mathbb{R}^{d\times d}$ is the parameter-space Gram matrix and the last step uses $\boldsymbol{G}_{\boldsymbol{\theta}} \succeq \mathbf{0}$ to establish proportionality via the cyclic property of the trace. ☐

D.2 Proof of Theorem 3 (Semi-Gradient Gap)

Theorem 6 (Semi-Gradient Gap)**.** *The gradients of the MeanFlow loss with and without the stop-gradient operator differ by*

$$\underbrace{2(t-r)\,\mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}[\nabla_{\boldsymbol{\theta}}(\partial_{\boldsymbol{x}_t}\boldsymbol{u}_{\boldsymbol{\theta}}\cdot\boldsymbol{v}(\boldsymbol{x}_t,t) + \partial_t\boldsymbol{u}_{\boldsymbol{\theta}})\cdot\boldsymbol{r}_{\boldsymbol{\theta}}]}_{\textit{mean-field gradient difference}} + \underbrace{\mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\big[\nabla_{\boldsymbol{\theta}}\,\mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big)\big]}_{\textit{variance-driven gradient difference}}.$$

Proof. The MeanFlow loss conditioned on $\boldsymbol{x}_t$ decomposes as (cf. equation 5):

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\ell_{\mathrm{MF}}] = \|\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathsf{sg}}\|^2 + \mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big)\,. \tag{17}$$

Without stop-gradient. Both terms depend on $\boldsymbol{\theta}$ (through $\boldsymbol{u}_{\boldsymbol{\theta}}$ and $\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}$), so the full gradient is:

$$\nabla_{\boldsymbol{\theta}}\mathbb{E}[\ell] = \nabla_{\boldsymbol{\theta}}\,\|\boldsymbol{r}_{\boldsymbol{\theta}}\|^2 + \nabla_{\boldsymbol{\theta}}\,\mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big)\,. \tag{18}$$

With stop-gradient. The JVP terms in $\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathsf{sg}}$ and $\boldsymbol{J}$ are treated as constants, so:

$$\nabla_{\boldsymbol{\theta}}^{\mathsf{sg}}\,\|\boldsymbol{r}^{\mathsf{sg}}\|^2 = 2\,(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top\,\boldsymbol{r}^{\mathsf{sg}}, \qquad \nabla_{\boldsymbol{\theta}}^{\mathsf{sg}}\,\mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big) = \mathbf{0}. \tag{19}$$

The gap between the full and stop-gradient gradients is therefore:

$$\begin{aligned}
&\nabla_{\boldsymbol{\theta}}\mathbb{E}[\ell] - \nabla_{\boldsymbol{\theta}}^{\mathsf{sg}}\mathbb{E}[\ell] \\
&\quad= \underbrace{\nabla_{\boldsymbol{\theta}}\,\|\boldsymbol{r}_{\boldsymbol{\theta}}\|^2 - 2\,(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top\,\boldsymbol{r}^{\mathsf{sg}}}_{\text{mean-field gradient difference}} + \underbrace{\nabla_{\boldsymbol{\theta}}\,\mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big)}_{\text{variance-driven difference}}.
\end{aligned} \tag{20}$$

Expanding the first term: $\nabla_{\boldsymbol{\theta}}\|\boldsymbol{r}\|^2 = 2(\nabla_{\boldsymbol{\theta}}\boldsymbol{r}_{\boldsymbol{\theta}})^\top\boldsymbol{r}_{\boldsymbol{\theta}}$, and $\nabla_{\boldsymbol{\theta}}\boldsymbol{r}_{\boldsymbol{\theta}}$ includes the derivative of the JVP terms $(t-r)(\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}\cdot\boldsymbol{v} + \partial_t\boldsymbol{u}_{\boldsymbol{\theta}})$ w.r.t. $\boldsymbol{\theta}$, yielding the second-order mean-field difference $2(t-r)\mathbb{E}[\nabla_{\boldsymbol{\theta}}(\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}\cdot\boldsymbol{v} + \partial_t\boldsymbol{u}_{\boldsymbol{\theta}})\cdot\boldsymbol{r}_{\boldsymbol{\theta}}]$. At convergence $\boldsymbol{r}_{\boldsymbol{\theta}} \to \mathbf{0}$, this term vanishes, and the gap is dominated by the variance-driven term $\nabla_{\boldsymbol{\theta}}\,\mathrm{Tr}(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top)$. ☐

D.3 Proof of Proposition 2 (Gradient of VaMF Loss)

Proposition 7 (Gradient of VaMF Loss)**.** *The gradient of $\mathcal{L}_{MF}^{EMA}$ takes the form*

$$\nabla_{\boldsymbol{\theta}}\mathcal{L}_{MF}^{EMA} \;=\; 2\,\mathbb{E}\big[\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t,r,t)\cdot\tilde{\boldsymbol{r}}^{EMA}\big]\,,$$

where $\tilde{\boldsymbol{r}}^{EMA} := V_{\boldsymbol{\theta}}^{EMA} - \boldsymbol{v}(\boldsymbol{x}_t,t)$ is the mean-field residual under the EMA tangent, and no $\boldsymbol{J}\boldsymbol{v}'$ noise term appears.

Proof. The EMA velocity anchor $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t,t,t)$ is deterministic given $\boldsymbol{x}_t$, since the EMA parameters $\bar{\boldsymbol{\theta}}$ are fixed during the gradient step. Under stop-gradient, $V_{\boldsymbol{\theta}}^{\mathrm{EMA}}$ depends on $\boldsymbol{\theta}$ only through $\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t,r,t)$. By the same argument as equation 14, the per-step gradient is:

$$\nabla_{\boldsymbol{\theta}}\ell = 2\,(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top\big(V_{\boldsymbol{\theta}}^{\mathrm{EMA}} - \boldsymbol{v}_{\mathrm{cond}}\big)\,. \tag{21}$$

Writing $\boldsymbol{v}_{\mathrm{cond}} = \boldsymbol{v}(\boldsymbol{x}_t,t) + \boldsymbol{v}'$ and noting that $\tilde{\boldsymbol{r}}^{\mathrm{EMA}} := V_{\boldsymbol{\theta}}^{\mathrm{EMA}} - \boldsymbol{v}(\boldsymbol{x}_t,t)$ is deterministic given $\boldsymbol{x}_t$:

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\nabla_{\boldsymbol{\theta}}\ell] = 2\,(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top\,\tilde{\boldsymbol{r}}^{\mathrm{EMA}}, \tag{22}$$

since $\mathbb{E}[\boldsymbol{v}' \mid \boldsymbol{x}_t] = \mathbf{0}$. The tower property $\nabla_{\boldsymbol{\theta}}\mathcal{L}_{\mathrm{MF}}^{\mathrm{EMA}} = \mathbb{E}_{r,t,\boldsymbol{x}_t}\big[\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\nabla_{\boldsymbol{\theta}}\ell]\big]$ gives the stated result. Crucially, no $\boldsymbol{J}\boldsymbol{v}'$ term appears because the EMA tangent is deterministic—compare with the original MeanFlow gradient equation 14 where the stochastic tangent produces the $\boldsymbol{J}\boldsymbol{v}'$ noise. ☐

D.4 Proof of Proposition 3 (Bias-Variance Tradeoff)

Proposition 8 (Bias-Variance Tradeoff)**.** *The EMA mean-field residual decomposes as*

$$\tilde{\boldsymbol{r}}^{EMA} \;=\; \boldsymbol{r}_{\boldsymbol{\theta}} \;+\; (t-r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t,t,t) - \boldsymbol{v}(\boldsymbol{x}_t,t))\,,$$

where $\boldsymbol{r}_{\boldsymbol{\theta}}$ is the true MeanFlow residual from equation 5. At convergence $\boldsymbol{r}_{\boldsymbol{\theta}} \to \mathbf{0}$ and the FM anchor drives $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t,t,t) \to \boldsymbol{v}(\boldsymbol{x}_t,t)$, giving $\tilde{\boldsymbol{r}}^{EMA} \to \mathbf{0}$.

Proof. Expanding $V_{\boldsymbol{\theta}}^{\text{EMA}}$ from equation 11, the value of the compound prediction (ignoring the stop-gradient, which does not affect values) is:

$$V_{\boldsymbol{\theta}}^{\text{EMA}} = \boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)(\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t,t,t) + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}})\,. \tag{23}$$

The true MeanFlow residual (with the marginal velocity as tangent) is:

$$\boldsymbol{r}_{\boldsymbol{\theta}} = \boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)(\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v}(\boldsymbol{x}_t,t) + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}) - \boldsymbol{v}(\boldsymbol{x}_t,t). \tag{24}$$

Subtracting:

$$\begin{aligned} \tilde{\boldsymbol{r}}^{\text{EMA}} &= V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}(\boldsymbol{x}_t,t) \\ &= \boldsymbol{r}_{\boldsymbol{\theta}} + (t-r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t,t,t) - \boldsymbol{v}(\boldsymbol{x}_t,t))\,. \end{aligned} \tag{25}$$

At convergence, the true residual $\boldsymbol{r}_{\boldsymbol{\theta}} \to \mathbf{0}$ (the MeanFlow identity is satisfied) and the FM anchor loss equation 12 supervises the boundary condition, driving $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t,t,t) \to \boldsymbol{v}(\boldsymbol{x}_t,t)$ and hence $\tilde{\boldsymbol{r}}^{\text{EMA}} \to \mathbf{0}$. ☐

D.5 Proof of Proposition 4 (Target–Tangent Bias Asymmetry)

Proposition 9 (Target–Tangent Bias Asymmetry)**.** *Let $\hat{\boldsymbol{v}} = \boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t,t,t)$ be a deterministic proxy with bias $\boldsymbol{b} \triangleq \hat{\boldsymbol{v}} - \boldsymbol{v}(\boldsymbol{x}_t,t)$. For the tangent-replacement loss (VaMF), the loss minimum $\boldsymbol{u}^\star$ satisfies $\boldsymbol{u}^\star(\boldsymbol{x}_t,r,t) - \boldsymbol{v}(\boldsymbol{x}_t,t) = -(t-r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}^\star\,\boldsymbol{b} + \mathcal{O}(\|\boldsymbol{b}\|^2)$. For the target-replacement loss, the loss minimum at $r = t$ satisfies $\boldsymbol{u}^\star(\boldsymbol{x}_t,t,t) - \boldsymbol{v}(\boldsymbol{x}_t,t) = \boldsymbol{b}$.*

Proof. **Tangent replacement.** Substituting $\hat{\boldsymbol{v}}$ for $\boldsymbol{v}_{\text{cond}}$ only in the JVP tangent (target unchanged) gives the per-sample compound prediction

$$V_{\boldsymbol{\theta}}^{\text{EMA}} = \boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)(\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{v}} + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}})\,,$$

and the loss is $\mathbb{E}\|V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}_{\text{cond}}\|^2$. By stationarity, $\mathbb{E}[V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}_{\text{cond}} \mid \boldsymbol{x}_t] = \mathbf{0}$ at the minimum. Taking conditional expectation and using $\mathbb{E}[\boldsymbol{v}_{\text{cond}} \mid \boldsymbol{x}_t] = \boldsymbol{v}(\boldsymbol{x}_t,t)$,

$$\boldsymbol{u}^\star + (t-r)(\partial_{\boldsymbol{z}}\boldsymbol{u}^\star \cdot \hat{\boldsymbol{v}} + \partial_t \boldsymbol{u}^\star) = \boldsymbol{v}(\boldsymbol{x}_t,t).$$

At the same point, the *true* MeanFlow identity (with the marginal velocity as tangent) reads

$$\boldsymbol{u}^\diamond + (t-r)(\partial_{\boldsymbol{z}}\boldsymbol{u}^\diamond \cdot \boldsymbol{v} + \partial_t \boldsymbol{u}^\diamond) = \boldsymbol{v}(\boldsymbol{x}_t,t),$$

with $\boldsymbol{u}^\diamond(\boldsymbol{x}_t,t,t) = \boldsymbol{v}(\boldsymbol{x}_t,t)$. Subtracting and writing $\boldsymbol{u}^\star = \boldsymbol{u}^\diamond + \Delta$,

$$\Delta + (t-r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}^\star\,(\hat{\boldsymbol{v}} - \boldsymbol{v}) + (t-r)\,\partial_t\Delta + (t-r)\,(\partial_{\boldsymbol{z}}\Delta)\,\boldsymbol{v} = \mathbf{0}.$$

Evaluating at $r = t$ gives $\Delta(\boldsymbol{x}_t,t,t) = \mathbf{0}$ (boundary condition recovered exactly). For $r < t$, the leading term in $\boldsymbol{b} = \hat{\boldsymbol{v}} - \boldsymbol{v}$ gives

$$\Delta(\boldsymbol{x}_t,r,t) = -(t-r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}^\star\,\boldsymbol{b} + \mathcal{O}(\|\boldsymbol{b}\|^2),$$

i.e., the stationary error is $\mathcal{O}((t-r)\,\|\partial_{\boldsymbol{z}}\boldsymbol{u}\|\,\|\boldsymbol{b}\|)$ and vanishes at the boundary $r \to t$.

Target replacement. Substituting $\hat{\boldsymbol{v}}$ for $\boldsymbol{v}_{\text{cond}}$ in the regression target yields the loss $\mathbb{E}\|\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\,\mathrm{D}\boldsymbol{u}/\mathrm{D}t - \hat{\boldsymbol{v}}\|^2$. By stationarity, the loss minimum satisfies $\mathbb{E}[\boldsymbol{u}^\star + (t-r)\,\mathrm{D}\boldsymbol{u}^\star/\mathrm{D}t - \hat{\boldsymbol{v}} \mid \boldsymbol{x}_t] = \mathbf{0}$. Specializing to $r = t$, the gap term vanishes and the equation collapses to $\boldsymbol{u}^\star(\boldsymbol{x}_t,t,t) = \hat{\boldsymbol{v}} = \boldsymbol{v}(\boldsymbol{x}_t,t) + \boldsymbol{b}$, so the stationary error at the boundary is exactly $\boldsymbol{b}$, independent of (t,r).

Asymmetry. Tangent replacement carries a multiplicative $(t-r)$ factor on the bias and is therefore pinned to zero at the boundary $r = t$ that the FM anchor enforces; target replacement carries no such factor and the bias persists at every t, requiring a boundary regularizer at *every* (r,t) to remove. ☐

D.6 Proof of Theorem 4 (Optimal Control-Variate Coefficient)

Theorem 10 (Optimal control-variate coefficient)**.** *Under the scalar-isotropic approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma^2 \boldsymbol{I}_d$ and $\boldsymbol{J} \approx \kappa \boldsymbol{I}_d$, the conditional MSE of $\boldsymbol{g}^{(\beta)}$ in equation 7 is*

$$M(\beta) \;=\; \beta^2(\kappa{+}1)^2\|\boldsymbol{b}\|^2 \;+\; \sigma^2 d\big((1{-}\beta)\kappa - \beta\big)^2,$$

with unique minimizer $\beta^\star = \frac{\kappa}{\kappa+1}\cdot\frac{\sigma^2 d}{\sigma^2 d+\|\boldsymbol{b}\|^2}$ and optimum value $M(\beta^\star) = \sigma^2 d\,\kappa^2\,\|\boldsymbol{b}\|^2/(\sigma^2 d + \|\boldsymbol{b}\|^2)$. For all κ, $\|\boldsymbol{b}\|^2 > 0$, the optimum strictly dominates both $M(0) = \sigma^2\kappa^2 d$ and $M(1) = (\kappa{+}1)^2\|\boldsymbol{b}\|^2 + \sigma^2 d$.

Proof. **Step 1: bias-variance decomposition.** From equation 7,

$$\boldsymbol{g}^{(\beta)} \;=\; 2\,\boldsymbol{g}^\top\Big[\bar{\boldsymbol{r}} \;+\; \big((1{-}\beta)\,\boldsymbol{J} - \beta\,\boldsymbol{I}\big)\,\boldsymbol{v}' \;+\; \beta\,(\boldsymbol{J}{+}\boldsymbol{I})\,\boldsymbol{b}\Big].$$

Using $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}'] = \boldsymbol{0}$, the conditional mean is $\mathbb{E}[\boldsymbol{g}^{(\beta)} \mid \boldsymbol{x}_t] = 2\boldsymbol{g}^\top(\bar{\boldsymbol{r}} + \beta(\boldsymbol{J}{+}\boldsymbol{I})\boldsymbol{b})$ and the centered noise is $-2\boldsymbol{g}^\top\big((1{-}\beta)\boldsymbol{J} - \beta\boldsymbol{I}\big)\boldsymbol{v}'$ (relative to the ideal target $2\boldsymbol{g}^\top\bar{\boldsymbol{r}}$). The conditional MSE w.r.t. the ideal target therefore decomposes as

$$\mathrm{MSE} \;=\; \underbrace{4\beta^2\,\big\|\boldsymbol{g}^\top(\boldsymbol{J}{+}\boldsymbol{I})\boldsymbol{b}\big\|^2}_{\text{bias}^2} \;+\; \underbrace{4\,\mathrm{Tr}\Big(\boldsymbol{g}\boldsymbol{g}^\top\big((1{-}\beta)\boldsymbol{J} - \beta\boldsymbol{I}\big)\Sigma_{\boldsymbol{v}'}\big((1{-}\beta)\boldsymbol{J} - \beta\boldsymbol{I}\big)^\top\Big)}_{\text{variance}}.$$

Step 2: scalar approximation. Substituting $\boldsymbol{J} = \kappa\boldsymbol{I}_d$ and $\Sigma_{\boldsymbol{v}'} = \sigma^2\boldsymbol{I}_d$, and absorbing the parameter-direction normalization $\mathrm{Tr}(\boldsymbol{g}\boldsymbol{g}^\top) = \|\boldsymbol{g}\|_F^2$ into a per-component MSE,

$$M(\beta) \;\propto\; \beta^2(\kappa{+}1)^2\|\boldsymbol{b}\|^2 \;+\; \sigma^2 d\big((1{-}\beta)\kappa - \beta\big)^2.$$

Step 3: minimization. Differentiating and setting to zero,

$$\begin{aligned}\frac{\partial M}{\partial\beta} &= 2\beta(\kappa{+}1)^2\|\boldsymbol{b}\|^2 - 2(\kappa{+}1)\sigma^2 d\big((1{-}\beta)\kappa - \beta\big) \;=\; 0\\ \iff\quad & \beta(\kappa{+}1)\big(\sigma^2 d + \|\boldsymbol{b}\|^2\big) \;=\; \sigma^2 d\,\kappa.\end{aligned}$$

Solving yields $\beta^\star = \sigma^2 d\,\kappa/[(\kappa{+}1)(\sigma^2 d + \|\boldsymbol{b}\|^2)]$, which factors as the noise-cancellation $\kappa/(\kappa{+}1)$ times the James-Stein shrinkage $\sigma^2 d/(\sigma^2 d + \|\boldsymbol{b}\|^2)$.

Step 4: optimum value. At $\beta^\star$, $(1{-}\beta^\star)\kappa - \beta^\star = \kappa\,\|\boldsymbol{b}\|^2/(\sigma^2 d + \|\boldsymbol{b}\|^2)$. Substituting back,

$$M(\beta^\star) \;=\; \frac{\sigma^4 d^2\kappa^2\|\boldsymbol{b}\|^2 + \sigma^2 d\,\kappa^2\|\boldsymbol{b}\|^4}{(\sigma^2 d + \|\boldsymbol{b}\|^2)^2} \;=\; \frac{\sigma^2 d\,\kappa^2\,\|\boldsymbol{b}\|^2}{\sigma^2 d + \|\boldsymbol{b}\|^2}.$$

Step 5: strict dominance over corners. For κ, $\|\boldsymbol{b}\|^2 > 0$: $M(\beta^\star)/M(0) = \|\boldsymbol{b}\|^2/(\sigma^2 d + \|\boldsymbol{b}\|^2) < 1$, so $M(\beta^\star) < M(0)$. For the $M(1)$ comparison, $M(1) - M(\beta^\star) = (\kappa{+}1)^2\|\boldsymbol{b}\|^2 + \sigma^2 d - \sigma^2 d\kappa^2\|\boldsymbol{b}\|^2/(\sigma^2 d + \|\boldsymbol{b}\|^2)$. The first two terms exceed $\sigma^2 d\kappa^2\|\boldsymbol{b}\|^2/(\sigma^2 d + \|\boldsymbol{b}\|^2)$ since $(\kappa{+}1)^2 > \kappa^2$ for $\kappa > 0$. ☐

E DiT Latent-Space Training on ImageNet

We test whether the variance-reduction mechanism of Theorems 2 and 4 scales beyond toys by training DiT [47] variants on ImageNet in the latent space of a frozen Stable Diffusion VAE, following the recipe of [21]. Both baselines and VaMF-L_2 runs share identical hyperparameters and code paths; only the loss differs (vanilla MeanFlow MSE versus the EMA-tangent VaMF-L_2 loss in equation 13 with a single Hutchinson probe).

E.1 DiT-B/4, ImageNet-256×256

Setup: logit-normal (r, t) sampling with mean -0.4 and stddev 1, overlap rate 0.75, AdamW with constant learning rate 10^{-4}, EMA decay 0.9999, batch size 256 across 32 TPU v4 chips. The single-probe Hutchinson estimator adds approximately one additional JVP per step, which we measure as a $\sim 22\%$ wall-clock overhead on TPU v4 (2.20 steps/s for VaMF-L_2 versus 2.65 steps/s for the baseline). All three configurations—baseline ($\beta = 0$, MeanFlow), interior probe ($\beta = 0.5$), and VaMF-L_2 ($\beta = 1$)—share the identical model, data, optimizer, and EMA schedule; only the tangent coefficient differs.

Three-floor structure and matched-step ordering. Table 3 reports FID_{50k} at representative 5k-step checkpoints for all three runs, and Figure 5 renders the full trajectories on a semilog-y axis. After the early-training transient (≤ 10k steps), the empirical ordering $\mathrm{FID}(\beta{=}0) < \mathrm{FID}(\beta{=}0.5) < \mathrm{FID}(\beta{=}1)$ holds at *every* matched-step checkpoint we logged (24 consecutive datapoints from step 30k onward). The trajectory has two phases: in the first (20–45k), the gap between $\beta = 0.5$ and the $\beta = 1$ corner widens monotonically (from $+2.1$ at step 20k to -11.4 at step 45k), validating the interior optimum predicted by Theorem 4; once the gradient-noise reduction of $\beta = 0.5$ has paid off, the convergence-rate advantage stabilizes. In the second phase (≥ 90k), the gap between $\beta = 0.5$ and the $\beta = 0$ baseline reaches an asymptotic offset of $\boxed{+7.77 \pm 0.11}$ FID over 11 consecutive matched-step windows (step 95k–145k), well below the $+16$ FID gap of the early-training phase.

$\beta = 0.5$ beats $\beta = 1$'s asymptotic floor. VaMF-L_2 ($\beta = 1$) completed its scheduled 300k-step training run with a final logged FID of $\mathbf{23.36}$ (step 295k); we treat this as the empirical $\beta = 1$ floor. The $\beta = 0.5$ run *passes* this floor at step 115k (FID 22.18) and continues descending; at the latest logged checkpoint (step 145k) it sits at FID 18.07, a margin of 5.3 FID below VaMF-L_2's asymptote with the run still in progress. The three converged FID floors recover the bias–variance ordering of Theorem 4:

$$\mathrm{FID}^\star(\beta{=}0) \;\approx\; 12.3 \;<\; \mathrm{FID}^\star(\beta{=}0.5) \;\approx\; 18\text{–}19 \;<\; \mathrm{FID}^\star(\beta{=}1) \;=\; 23.4,$$

with the $\beta = 0.5$ floor still descending at the cut-off. The bias amplification ratio in FID across $\beta = 0.5/\beta = 1$ is approximately 1.4:1, not the 1:4 that the gradient-MSE bias term $\beta^2(\kappa{+}1)^2\|\boldsymbol{b}\|^2$ predicts—a quantitative measure of the FID-vs-MSE landscape mismatch discussed in Section 5.5. The three-floor structure is the strongest empirical confirmation of the bias–variance decomposition we obtain at DiT scale.

Table 3: DiT-B/4 / ImageNet-256 FID_{50k} (lower is better) at representative 5k-step checkpoints for the baseline ($\beta = 0$, wandb `hde9iaqj`, finished step 235k), the interior probe ($\beta = 0.5$, wandb `milo2x6t`, still running), and VaMF-L_2 ($\beta = 1$, wandb `hrnr0uvz`, finished step 295k). Bottom rows give the matched-step deltas. Note that $\beta = 0.5$ at step 145k (FID 18.07) already sits 5.3 FID below VaMF-L_2's final FID 23.36, with $\beta = 0.5$ still descending.

step (k)	30	40	50	60	80	100	115	130	145	final
baseline ($\beta{=}0$)	128.9	79.2	54.5	41.8	28.3	17.7	14.4	12.1	10.3	12.1†
$\beta{=}0.5$	145.2	94.6	—	—	—	25.5	22.2	19.7	18.1	(running)
VaMF-L_2 ($\beta{=}1$)	148.8	104.1	79.9	65.7	49.9	41.7	38.0	35.2	32.7	**23.4**
Δ ($\beta{=}0.5$ − baseline)	+16.2	+15.5	—	—	—	+7.9	+7.8	+7.6	+7.8	—
Δ ($\beta{=}0.5$ − VaMF-L_2)	−3.6	−9.4	—	—	—	−16.2	−15.8	−15.5	−14.6	—

† Baseline run logged its last eval at step 235k with FID 12.12 before being stopped. We extrapolate the floor as $\mathrm{FID}^\star \approx 12.3$.

Bias and shrinkage trajectory. To localize the predicted optimum we measure both ingredients of Theorem 4 directly on the DiT checkpoints: the EMA boundary-tracking bias by $\|\boldsymbol{b}\|^2 \triangleq \mathbb{E}_{\boldsymbol{x}_t}\left[\|\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t, t) - \boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t)\|^2\right]$ at synthetic-Gaussian $\boldsymbol{x}_t$, and the irreducible noise floor $\sigma^2 d \approx 8.2{\times}10^3$ from $\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})$ on the same batch. Table 4 reports the trajectory: the bias *decays* through training, and the shrinkage factor $\sigma^2 d/(\sigma^2 d{+}\|\boldsymbol{b}\|^2)$ stays near 1 throughout converged training. Combined with $\kappa \approx 1$ from the Hutchinson trace, this places $\beta^\star \in [0.46, 0.50]$ across the full trajectory—firmly interior in the gradient-MSE landscape.

Table 4: Direct DiT-checkpoint measurement of the bias-variance decomposition ingredients of Theorem 4 at $t = 0.5$. The EMA bias $\|\boldsymbol{b}\|^2$ averages over 128 synthetic-Gaussian $\boldsymbol{x}_t$ samples and is reported for both methods (which track within 20% of each other). The shrinkage factor uses $\sigma^2 d \approx 8.2 \times 10^3$. The predicted $\beta^\star$ uses $\kappa = 1$ from the Hutchinson trace.

step (k)	$\|\boldsymbol{b}\|^2$ (avg)	$\sigma^2 d/(\sigma^2 d + \|\boldsymbol{b}\|^2)$	$\beta^\star$
20	~ 700	0.92	0.46
40	~ 150	0.98	0.49
80	~ 50	0.99	0.50